Lab 1: Requirements Engineering & UML Use-Case Modelling

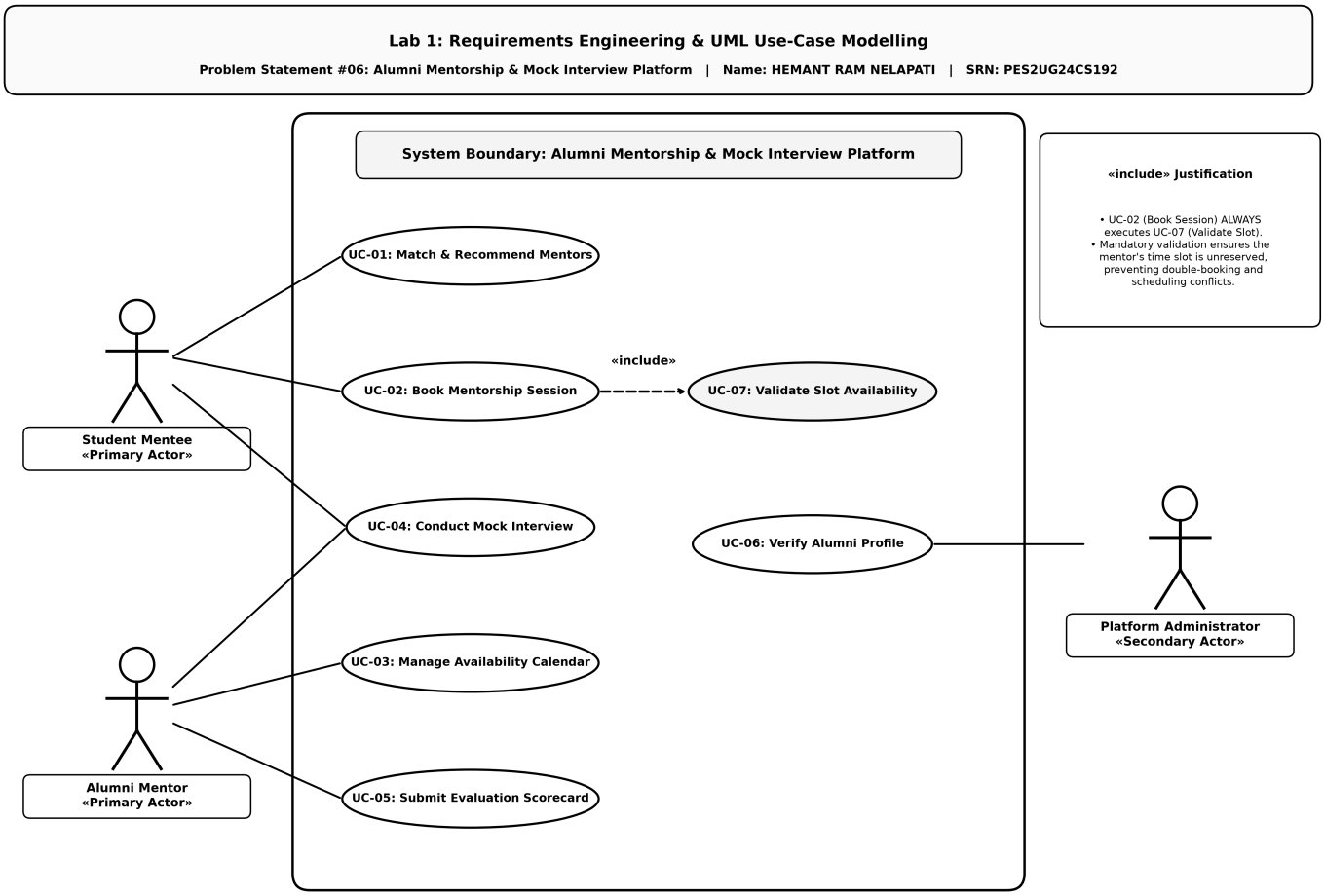
Problem Statement #06: Alumni Mentorship & Mock Interview Platform (Campus & Academic Operations)

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1. Requirements Specification Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall recommend verified alumni mentors to student mentees based on their selected domain specialization (e.g., Embedded Systems, Cloud Computing).	**High**	Pass: When a student selects a domain (e.g., "Cloud Computing"), the system displays verified alumni mentors associated with that domain, with mentors having the strongest domain match presented first. Fail: Unverified or domain-unmatched mentors are recommended.	Core mentor discovery mechanism connecting students with relevant alumni expertise for targeted career preparation.	Domain tags are selected from pre-approved campus academic taxonomies.
FR-002	Functional	The system shall book a mock interview session for a student mentee from the mentor's published open calendar slots and prevent double-booking.	**High**	Pass: Selected time slot is confirmed, added to both student and mentor platform schedules, and locked from further booking. Fail: Double-booking occurs on the same mentor slot.	Eliminates scheduling conflicts and provides dependable session booking.	Slot availability is validated in real-time before booking confirmation.
FR-003	Functional	The system shall allow alumni mentors to create, update, and publish their available dates and time slots for mentorship sessions.	**Medium**	Pass: Mentor-defined availability intervals immediately appear as selectable slots on the student booking interface. Fail: Published slots do not reflect on student calendar or past dates are allowed.	Enables alumni mentors to manage their contribution time according to their professional schedule.	Mentors can mark specific slots as unavailable prior to reservation.
FR-004	Functional	The system shall allow the alumni mentor to record and submit a structured post-interview scorecard using standardized evaluation rubrics and feedback.	**High**	Pass: Scorecard containing ratings across standardized rubrics (Technical Skills, Problem Solving, Communication) is saved and made accessible to the mentee. Fail: Submission allowed with mandatory evaluation rubrics omitted.	Delivers structured, actionable evaluation to help students improve interview performance.	Scorecard submission completes the mock interview session lifecycle.
FR-005	Functional	The system shall allow the platform administrator to review and verify alumni profiles and domain qualifications before mentor activation.	**Medium**	Pass: Alumni profile becomes visible in student mentor search only after administrator status is marked as 'Verified'. Fail: Unverified alumni profile appears in search or booking.	Preserves platform credibility and ensures high-quality alumni guidance.	Admin verification status is logged and maintained on the mentor profile.
NFR-001	Nonfunctional	The system shall restrict access to a student's post-interview evaluation scorecard to the concerned student, the corresponding alumni mentor, and authorized platform administrators.	**High**	Pass: Only the concerned student mentee, associated alumni mentor, and authorized platform administrator can retrieve the scorecard. Fail: Any unauthorized platform user can retrieve or view the scorecard.	Protects student performance data confidentiality and ensures privacy of academic feedback.	Enforced via platform role-based access control (RBAC).
NFR-002	Nonfunctional	The system shall return domain-matched mentor search results within 2.0 seconds under simulated peak campus placement load.	**Medium**	Pass: 95% of mentor search queries return complete matching results in <= 2.0 seconds during peak concurrent student access. Fail: Search response time exceeds 2.0 seconds.	Maintains responsive platform performance during high-traffic academic placement and mock interview periods.	Evaluated through standard performance benchmarking.

2. UML Use-Case Diagram



Model Summary & Stereotype Justification	• **Actors:** Student Mentee (Primary), Alumni Mentor (Primary), Platform Administrator (Secondary) • **Mandatory Include:** UC-02 (Book Mentorship Session) «include» UC-07 (Validate Slot Availability) ensures that every booking unconditionally checks slot availability to prevent double-booking.

3. Use-Case Flow Specification

Use Case ID & Name:	**UC-02: Book Mentorship Session**
Primary Actor:	Student Mentee
Secondary Actor:	None
Preconditions:	1. Student Mentee is logged into the platform. 2. Alumni Mentor has published active availability slots.
Postconditions:	1. Mock interview session is confirmed and added to both platform schedules. 2. Selected slot is locked to prevent double-booking.

Main Success Scenario (MSS)

1. Student Mentee browses recommended mentors and opens a verified Alumni Mentor profile.

2. System retrieves and displays the mentor's profile details and published availability calendar.

3. Student Mentee selects an available date and time slot for a mock interview.

4. System executes **UC-07: Validate Slot Availability** (*«include»*) to verify the selected slot remains unreserved.

5. System creates the session reservation and locks the time slot to prevent double-booking.

6. System adds the confirmed session to both the Student Mentee's and Alumni Mentor's platform schedules.

7. System displays a booking confirmation message with complete session details to the Student Mentee.

Alternate Flow

4a. Selected Slot Unavailable / Concurrently Booked (Branch from Step 4)

4a1. System detects that the selected time slot was locked by another student or is no longer open.

4a2. System displays an alert notification: *'The selected time slot is no longer available.'*

4a3. System refreshes the mentor's availability calendar with remaining open slots.

4a4. Student Mentee selects an alternative open time slot (resumes at Step 3) or cancels booking.